Table 5. Overview of fair stream learning algorithms

		ML Algorithm	Fairness target metric	Fairness pre-/post/in	Drift	Imbalance	Code published
pre	Massaging [14]	model agnostic	DP	pre	✗	✗	✓
pre	Reweighing [14]	model agnostic	DP	pre	✗	✗	✓
pre	CFSMOTE [18]	model agnostic	individual	pre	✓	✓	✓
tree	FAHT [31]	Hoeffding Tree	DP	in	✗	✗	✓
tree	FEAT [29]	HAT	DP	in	✓	✗	✗
tree	2FAHT [32]	CVFDT	DP	in	✓	✗	✗
tree	FARF [30]	ARF	DP	in	✓	✗	✗
evolutionary	FOMOS [24]	Hoeffding Tree	DP	pre/in	✗	✗	✗
evolutionary	FAS Stream [23]	SAM, DSOS	DP	pre/in	✓	✗	✗
evolutionary	EMO-SAM [1]	SAM-kNN	DP	pre/in	✓	✗	✗
evolutionary	Fair-CMNB [2]	Mixed Naive Bayes		in/post	✓	✓	✗
post	FABBOO [13]	OzaBoost (model agnostic)	DP/EqOp/ EqFPR	post	✓	✓	✓

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	3.94 ± 0.52	35.5 ± 3.58	4.27 ± 1.14	**1.61 ± 0.34**	3.64 ± 0.9	80.4 ± 0.19	62.3 ± 0.4	28.5 ± 0.95
FEAT	6.26 ± 1.65	74.7 ± 18.8	13.7 ± 4.01	3.4 ± 0.87	6.45 ± 1.47	78 ± 0.31	55.5 ± 1.04	13.7 ± 2.4
2FAHT	4.34 ± 0.75	36.1 ± 2.2	5.69 ± 2.07	1.84 ± 0.35	**0.57 ± 0.05**	80.7 ± 0.31	63.4 ± 1.23	31.2 ± 2.98
CFSMOTE	10.5 ± 0.88	28.2 ± 2.56	6.79 ± 1.55	6.25 ± 0.73	6.81 ± 0.74	78.5 ± 0.28	76.9 ± 0.15	73.9 ± 0.78
MAS	9.37 ± 0.54	36.5 ± 1.7	11.7 ± 1.52	4.74 ± 0.45	3.41 ± 0.4	**82.8 ± 0.38**	72.5 ± 0.24	53.3 ± 0.59
FABBOO SP	**0.44 ± 0.24**	**1.39 ± 0.8**	5.51 ± 0.5	4.64 ± 0.2	16.6 ± 0.4	77.7 ± 0.24	77.2 ± 0.15	**76.4 ± 0.62**
FABBOO EQOP	8.11 ± 1.14	20.8 ± 3.13	**2.27 ± 0.8**	3.88 ± 1.18	10.6 ± 0.79	79.5 ± 0.26	**78.1 ± 0.23**	75.6 ± 0.68

Table 6. Fairness and performance comparison across metrics on the *normal and fair world switch* stream (1 drift) with drift-aware scoring.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	16.4 ± 1.15	64.1 ± 2.83	15.7 ± 1.76	8.02 ± 1.01	8.98 ± 1.96	76.7 ± 0.13	65.7 ± 0.22	39.7 ± 0.51
FEAT	26.6 ± 1.27	92.1 ± 1.21	36.6 ± 1.99	15.6 ± 1	17.1 ± 2.21	76.4 ± 0.56	64.9 ± 1.07	40 ± 2.55
2FAHT	23.1 ± 1.28	71.5 ± 2.41	23.4 ± 2.16	13.1 ± 0.92	**1.35 ± 0.34**	78.2 ± 0.17	68.8 ± 0.92	48.6 ± 2.43
CFSMOTE	20.5 ± 0.94	47.3 ± 1.77	10.4 ± 1.8	9.46 ± 0.82	6.56 ± 0.82	78.7 ± 0.35	77.2 ± 0.23	74.1 ± 0.78
MAS	23.5 ± 1.67	68.6 ± 1.35	23.1 ± 1.54	11.4 ± 1.77	4.96 ± 0.66	**80.4 ± 0.3**	74.5 ± 0.33	60.1 ± 2.23
FABBOO SP	**0.35 ± 0.1**	**0.74 ± 0.18**	9.47 ± 0.4	11.8 ± 0.25	30 ± 0.56	74.3 ± 0.32	76.2 ± 0.16	**82.1 ± 0.41**
FABBOO EQOP	16.8 ± 0.6	36 ± 1.58	**2.67 ± 0.76**	**5.76 ± 0.67**	16.8 ± 1.15	78.3 ± 0.34	**78.6 ± 0.16**	79.9 ± 0.49

Table 7. Fairness and performance comparison across metrics on the *inflation scenario* stream (1 drift) with drift-aware scoring.

A Additional Tables and Results

B Online Resources

The code can be found on GitHub.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	8.44 ± 1	61.7 ± 2.94	8.88 ± 1.91	**3.41 ± 0.66**	4.57 ± 1.1	79.7 ± 0.12	62.8 ± 0.55	30.1 ± 1.43
FEAT	10.2 ± 2.23	96.9 ± 1.81	21.1 ± 4.91	5.51 ± 1.12	8.06 ± 1.69	77.3 ± 0.53	57 ± 1.72	17.8 ± 4.09
2FAHT	8.56 ± 1.15	67.7 ± 4.09	10.6 ± 3.37	3.67 ± 0.65	**0.53 ± 0.16**	79.4 ± 0.28	61.8 ± 0.63	27.5 ± 1.59
CFSMO HAT	16.6 ± 2.19	36.7 ± 3.48	6.17 ± 2.67	9.42 ± 2.39	2.11 ± 0.42	73.5 ± 1.21	74.3 ± 0.65	75.9 ± 1
CFSMO ARF	17.2 ± 0.69	45.5 ± 1.69	9.37 ± 0.93	8.46 ± 0.76	7.89 ± 1.45	78.9 ± 0.34	76.7 ± 0.24	72.5 ± 0.6
MAS	17.8 ± 0.74	69.2 ± 1.92	19.7 ± 2.21	8.77 ± 0.61	4.74 ± 1	**82.4 ± 0.3**	73 ± 0.22	54.6 ± 0.63
FABBOO SP	**0.18 ± 0.06**	**0.46 ± 0.14**	10.1 ± 0.55	8.54 ± 0.16	29.5 ± 0.64	74.8 ± 0.29	76.2 ± 0.11	**79 ± 0.42**
FABBOO EQOP	13 ± 0.71	32.9 ± 1.73	**1.42 ± 0.6**	4.57 ± 0.8	18.3 ± 1.06	78.1 ± 0.3	**77.7 ± 0.14**	77 ± 0.48

Table 8. Fairness and performance comparison across metrics on the *women in stem scenario* stream with drift-aware scoring.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	8.87 ± 0.86	64 ± 2.33	9.78 ± 1.45	**3.58 ± 0.64**	4.67 ± 1.15	79.9 ± 0.21	63 ± 0.54	30.4 ± 1.43
FEAT	11.9 ± 1.96	93.5 ± 7.43	23.4 ± 5.66	6.18 ± 0.92	8.71 ± 2.21	78.2 ± 0.86	59.5 ± 2.31	23.3 ± 5.16
2FAHT	9.42 ± 1.16	68.2 ± 3.56	11.2 ± 2.49	4.02 ± 0.67	**0.72 ± 0.24**	79.8 ± 0.36	62.9 ± 1.02	30.1 ± 2.43
CFSMOTE	17.3 ± 0.34	44.9 ± 1.06	6.57 ± 1.29	8.93 ± 0.36	7.07 ± 0.96	78.8 ± 0.38	77.3 ± 0.26	74.3 ± 0.31
MAS	17.9 ± 0.65	69.7 ± 1.54	19.9 ± 1.45	8.68 ± 0.63	4.73 ± 0.99	**82.5 ± 0.33**	73 ± 0.22	54.6 ± 0.52
FABBOO SP	**0.21 ± 0.08**	**0.53 ± 0.19**	10.8 ± 0.73	8.69 ± 0.28	30.2 ± 1.02	74.8 ± 0.26	76.3 ± 0.11	**79.1 ± 0.52**
FABBOO EQOP	13.7 ± 0.86	34.9 ± 2	**1.38 ± 0.56**	5.2 ± 0.81	18.2 ± 0.8	78.3 ± 0.24	**77.9 ± 0.24**	77 ± 0.55

Table 9. Fairness and performance comparison across metrics on the *less marriage scenario* stream with drift-aware scoring.

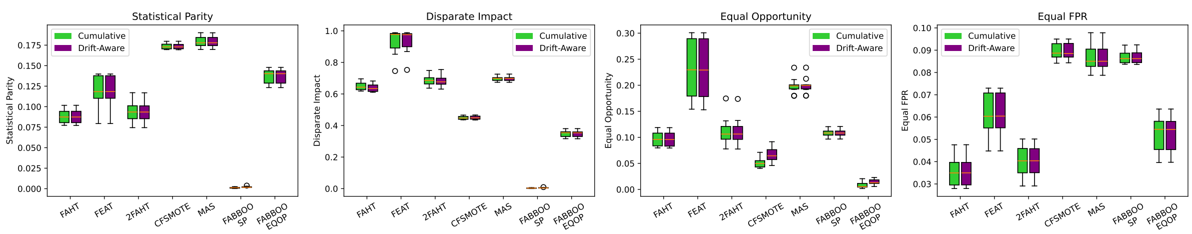

Fig. 11. Paired boxplots showing the differences between cumulatively and drift-awarely aggregated scores on the *less marriage scenario* stream with one drift.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	**0.26 ± 0.11**	**0.26 ± 0.11**	**0.27 ± 0.1**	**0.3 ± 0.31**	**0 ± 0.01**	86.3 ± 0.22	50.1 ± 0.23	**99.7 ± 0.13**
FEAT	1.26 ± 0.59	1.28 ± 0.6	0.81 ± 0.43	2.57 ± 1.24	0.1 ± 0.08	86.4 ± 0.28	52.5 ± 1.05	99 ± 0.36
2FAHT	0.51 ± 0.25	0.51 ± 0.25	0.44 ± 0.14	1.21 ± 0.65	0.02 ± 0.01	86.2 ± 0.35	50.5 ± 0.43	99.3 ± 0.28
CFSMOTE	6.45 ± 1.78	7.57 ± 2.03	5.56 ± 1.22	6.63 ± 3.15	6.39 ± 2.95	77.1 ± 2.14	57.2 ± 2.68	84.5 ± 2.48
MAS	2.55 ± 1.4	2.6 ± 1.43	1.58 ± 1.02	4.84 ± 2.45	0.34 ± 0.3	**86.4 ± 0.35**	53.7 ± 1.4	98.5 ± 0.66
EMOSAM	0.34 ± 0.23	0.34 ± 0.24	0.36 ± 0.28	0.48 ± 0.45	**0 ± 0**	86 ± 0.4	50 ± 0.27	99.3 ± 0.36
FABBOO SP	2.95 ± 0.92	3.22 ± 0.99	2.02 ± 0.74	9.82 ± 1.51	7.74 ± 0.46	84.8 ± 0.42	62.4 ± 0.91	93 ± 0.62
FABBOO EQOP	5.19 ± 1.41	5.67 ± 1.55	3.68 ± 0.99	10.4 ± 2.83	6.48 ± 0.58	84.6 ± 0.45	**63.4 ± 1.12**	92.4 ± 0.76

Table 10. Fairness and performance comparison across metrics on the *girls support* stream (three drifts) with drift-aware scoring.

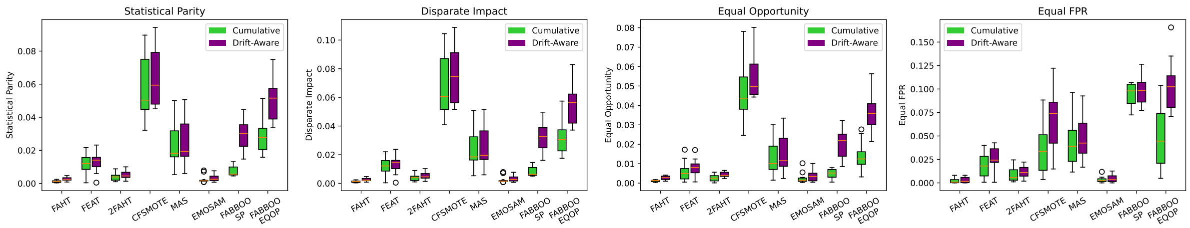

Fig. 12. Paired boxplots showing the differences between cumulatively and drift-awarely aggregated scores on the *girls support* stream with three drifts.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Acucracy	Recall
FAHT	**0.12** ± **0.07**	**0.12** ± **0.07**	**0.13** ± **0.09**	**0.2** ± **0.15**	0.01 ± 0.02	85.9 ± 0.32	50 ± 0.1	**99.8** ± **0.07**
FEAT	1.83 ± 0.53	1.87 ± 0.55	1.11 ± 0.34	2.51 ± 1.63	0.1 ± 0.1	**85.9** ± **0.42**	53 ± 1.04	98.8 ± 0.34
2FAHT	0.69 ± 0.39	0.7 ± 0.4	0.54 ± 0.33	1.61 ± 0.65	0.05 ± 0.04	85.7 ± 0.34	51.3 ± 1.04	99.1 ± 0.43
CFSMOTE	6.53 ± 1.37	7.65 ± 1.73	5.54 ± 1.13	6.75 ± 2.97	7.89 ± 4.1	76.7 ± 2.05	58.3 ± 1.69	83.8 ± 2.57
MAS	2.55 ± 1.09	2.63 ± 1.14	1.55 ± 0.81	3.98 ± 2.23	0.52 ± 0.37	85.9 ± 0.56	54.8 ± 1.72	98 ± 1.09
EMOSAM	0.33 ± 0.22	0.34 ± 0.22	0.3 ± 0.24	0.64 ± 0.39	**0** ± **0**	85.3 ± 0.62	49.9 ± 0.23	99 ± 0.52
FABBOO SP	2.45 ± 0.74	2.75 ± 0.81	1.35 ± 0.67	5.8 ± 2.29	4.96 ± 0.88	83.7 ± 0.68	63.8 ± 0.61	91.3 ± 1.04
FABBOO EQOP	4.29 ± 1.16	4.8 ± 1.29	2.66 ± 0.85	4.47 ± 1.07	4.1 ± 1.04	83.5 ± 0.69	**64.6** ± **0.53**	90.8 ± 1.09

Table 11. Fairness and performance comparison across metrics on the *girls support* stream (one drift) with drift-aware scoring.

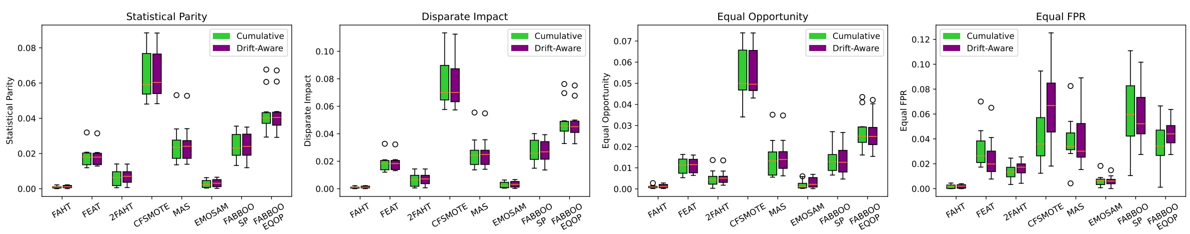

Fig. 13. Paired boxplots showing the differences between cumulatively and drift-awarely aggregated scores on the *girls support* stream with one drift.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Acucracy	Recall
FAHT	**0.16** ± **0.15**	**0.16** ± **0.15**	**0.16** ± **0.14**	**0.24** ± **0.22**	**0.01** ± **0.02**	85.3 ± 0.34	50 ± 0.14	**99.7** ± **0.11**
FEAT	0.74 ± 0.31	0.77 ± 0.33	0.52 ± 0.22	1.41 ± 0.95	0.14 ± 0.08	**85.4** ± **0.36**	53.1 ± 0.85	98.6 ± 0.37
2FAHT	0.42 ± 0.15	0.43 ± 0.16	0.34 ± 0.13	2.03 ± 0.98	0.07 ± 0.06	85.1 ± 0.38	51 ± 0.49	99.1 ± 0.39
CFSMOTE	2.6 ± 0.73	3.15 ± 0.9	2.28 ± 0.6	5.26 ± 2.29	5.9 ± 2.85	76.7 ± 1.58	59.3 ± 2.48	83.8 ± 1.79
MAS	1.09 ± 0.71	1.14 ± 0.77	0.76 ± 0.66	2.87 ± 1.18	0.44 ± 0.39	85.3 ± 0.33	55.3 ± 2.51	97.6 ± 1.23
FABBOO SP	2.07 ± 0.6	2.43 ± 0.7	1.64 ± 0.52	8.51 ± 1.96	2.65 ± 0.53	82.4 ± 0.6	66.5 ± 0.8	88.9 ± 0.75
FABBOO EQOP	2.74 ± 0.6	3.22 ± 0.71	2.06 ± 0.56	8.5 ± 2.83	2.6 ± 0.58	82.3 ± 0.55	**66.7** ± **0.64**	88.7 ± 0.7

Table 12. Fairness and performance comparison across metrics on the *boys support* stream (two drifts) with drift-aware scoring.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	**0.11 ± 0.06**	**0.11 ± 0.06**	**0.07 ± 0.04**	**0.34 ± 0.38**	0.03 ± 0.07	**85.7 ± 0.44**	50.1 ± 0.47	**99.7 ± 0.17**
FEAT	0.27 ± 0.3	0.28 ± 0.32	0.22 ± 0.2	1.67 ± 1.9	0.12 ± 0.15	**85.7 ± 0.45**	53 ± 2.14	98.7 ± 0.96
2FAHT	0.32 ± 0.17	0.32 ± 0.18	0.25 ± 0.12	1.27 ± 0.75	0.06 ± 0.04	85.6 ± 0.4	51 ± 0.58	99.3 ± 0.31
CFSMOTE	2.73 ± 1.15	3.27 ± 1.36	2.32 ± 1.1	7.24 ± 1.87	6.48 ± 3.16	77.2 ± 1.1	59.3 ± 2.39	84.3 ± 1.05
MAS	0.64 ± 0.56	0.67 ± 0.6	0.48 ± 0.41	2.69 ± 1.57	0.52 ± 0.35	85.6 ± 0.45	54.5 ± 2.44	97.9 ± 1.18
EMOSAM	0.32 ± 0.24	0.33 ± 0.25	0.29 ± 0.23	0.73 ± 0.52	**0 ± 0**	85 ± 1.13	49.9 ± 0.34	98.9 ± 1.35
FABBOO SP	1.12 ± 0.53	1.32 ± 0.62	0.88 ± 0.36	5.9 ± 2.7	2.26 ± 0.54	82.5 ± 0.73	66.5 ± 0.67	88.9 ± 0.93
FABBOO EQOP	1.8 ± 0.88	2.13 ± 1.04	1.37 ± 0.74	6.05 ± 2.42	2.37 ± 0.64	82.5 ± 0.73	**66.6 ± 0.58**	88.8 ± 0.95

Table 13. Fairness and performance comparison across metrics on the *boys support* stream (one drift) with drift-aware scoring.

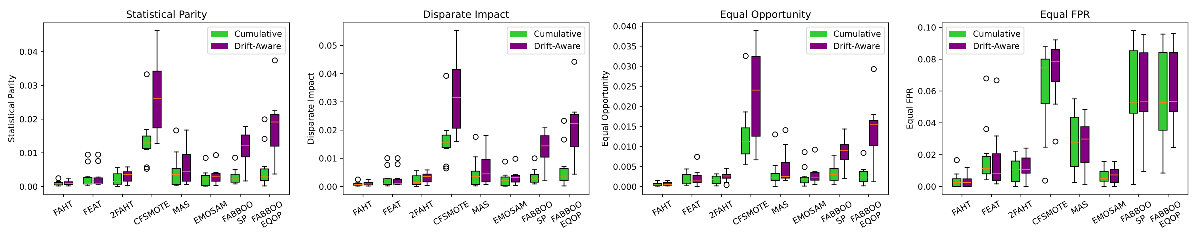

Fig. 14. Paired boxplots showing the differences between cumulatively and drift-awarely aggregated scores on the *boys support* stream with one drift.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	3.14 ± 1.05	5.46 ± 1.64	2.13 ± 1.78	3.03 ± 1.02	2.24 ± 0.82	64.5 ± 1	51.9 ± 0.26	75.7 ± 2.88
FEAT	1.06 ± 0.52	4.17 ± 2.14	1.5 ± 0.74	1.19 ± 0.65	0.54 ± 0.24	71.1 ± 0.95	50.8 ± 0.63	54.3 ± 2.72
2FAHT	1.06 ± 0.68	7.65 ± 4.09	**0.98 ± 0.69**	1.5 ± 1.04	0.46 ± 0.52	73.5 ± 1.34	51 ± 0.78	47.1 ± 4.59
CFSMOTE	2.05 ± 0.93	3.62 ± 1.9	3.08 ± 1.65	2.71 ± 1.18	7.83 ± 2.24	62.7 ± 1.21	54.8 ± 1.47	63.7 ± 1.85
MAS	1.38 ± 0.82	3.78 ± 2.24	1.5 ± 0.77	1.44 ± 0.97	0.36 ± 0.38	69 ± 0.74	51.8 ± 1.61	59.4 ± 1.4
EMOSAM	**0.58 ± 0.29**	8.49 ± 3.68	1.05 ± 0.43	**1.02 ± 0.59**	**0 ± 0**	**74.1 ± 0.97**	50 ± 0.31	42 ± 1.17
FABBOO SP	1.27 ± 0.51	**1.8 ± 0.81**	3.93 ± 1.82	3.53 ± 1.53	5.93 ± 0.65	64.7 ± 0.5	60.8 ± 0.4	**76 ± 1.16**
FABBOO EQOP	4.68 ± 2.07	7.54 ± 3.59	2.43 ± 0.89	5.49 ± 2.36	6.29 ± 1.26	65.2 ± 0.63	**61 ± 0.4**	74.6 ± 1.25

Table 14. Fairness and performance comparison across metrics on the *grade category adjustment* stream (one drift) with drift-aware scoring.

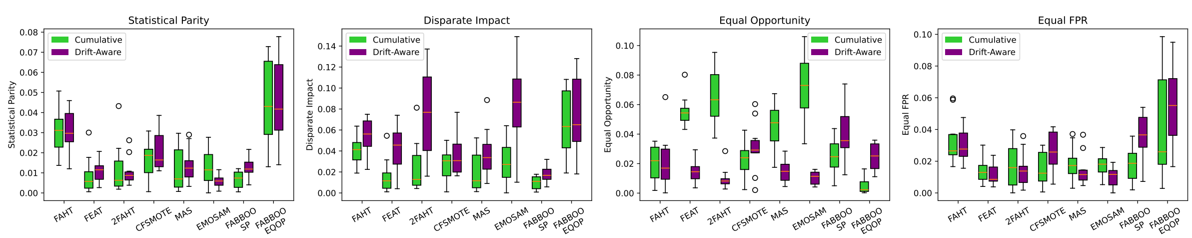

Fig. 15. Paired boxplots showing the differences between cumulatively and drift-awarely aggregated scores on the *grade category adjustment* stream with one drift.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	**0.11 ± 0.07**	**0.11 ± 0.07**	**0.14 ± 0.06**	**0.22 ± 0.24**	**0 ± 0.01**	**87.3 ± 0.28**	50 ± 0.07	**99.8 ± 0.08**
FEAT	0.6 ± 0.46	0.61 ± 0.47	0.46 ± 0.37	0.97 ± 0.83	0.06 ± 0.07	87.2 ± 0.32	51.7 ± 0.94	99.1 ± 0.38
2FAHT	0.41 ± 0.22	0.42 ± 0.23	0.33 ± 0.2	1.23 ± 0.61	0.04 ± 0.03	87.2 ± 0.38	51 ± 0.57	99.2 ± 0.35
CFSMOTE	2.74 ± 0.61	3.49 ± 0.73	2.01 ± 0.75	4.08 ± 2.88	5.48 ± 2.56	74.9 ± 3.2	59.7 ± 1.78	80.2 ± 4.14
MAS	1.24 ± 0.79	1.29 ± 0.84	1.1 ± 0.74	1.34 ± 1.19	0.54 ± 0.33	86.9 ± 0.5	53.8 ± 2.23	98 ± 1.23
EMOSAM	0.34 ± 0.23	0.34 ± 0.23	0.32 ± 0.17	0.77 ± 0.71	**0 ± 0**	87 ± 0.4	50 ± 0.2	99.3 ± 0.39
FABBOO SP	1.42 ± 0.46	1.64 ± 0.54	1.09 ± 0.54	7.25 ± 2.92	3.93 ± 0.68	83.7 ± 0.85	64.7 ± 0.75	89.9 ± 1.08
FABBOO EQOP	2.21 ± 0.97	2.57 ± 1.11	1.59 ± 0.61	6.04 ± 3	3.55 ± 0.93	83.5 ± 0.85	**65.1 ± 0.6**	89.6 ± 1.08

Table 15. Fairness and performance comparison across metrics on the *grade inflation* stream (one drift) with drift-aware scoring.

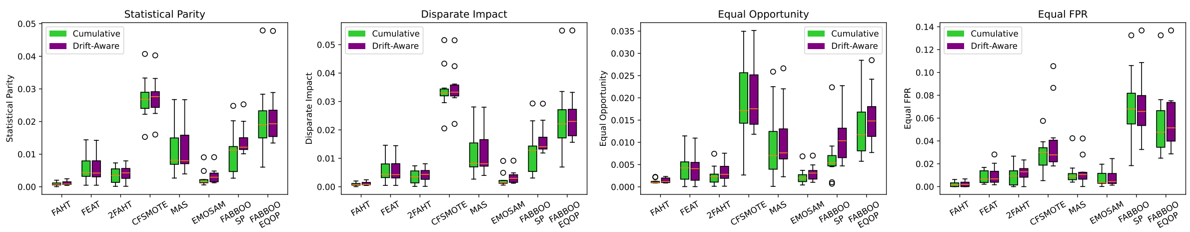

Fig. 16. Paired boxplots showing the differences between cumulatively and drift-awarely aggregated scores on the *grade inflation* stream with one drift.

Classifier	Statistical Parity	Disparate Impact	Equal Opportunity	Equal FPR	Individual Fairness	Accuracy	Balanced Accuarcy	Recall
FAHT	**0.15 ± 0.14**	**0.16 ± 0.14**	**0.17 ± 0.12**	**0.29 ± 0.24**	0.01 ± 0.02	85.1 ± 0.29	50 ± 0.09	**99.7 ± 0.1**
FEAT	0.7 ± 0.33	0.72 ± 0.34	0.54 ± 0.25	1.69 ± 1.18	0.16 ± 0.1	**85.4 ± 0.4**	53.3 ± 0.87	98.7 ± 0.36
2FAHT	0.34 ± 0.16	0.34 ± 0.16	0.21 ± 0.18	1.2 ± 0.63	0.05 ± 0.04	85.1 ± 0.35	51.1 ± 0.68	99.3 ± 0.32
CFSMOTE	3.08 ± 0.75	3.95 ± 0.92	2.03 ± 0.66	3.26 ± 1.57	5.07 ± 1.94	75.9 ± 2.53	64.3 ± 2.48	80.8 ± 2.63
MAS	1.08 ± 0.75	1.14 ± 0.8	0.79 ± 0.56	2.15 ± 1.05	0.59 ± 0.4	85.3 ± 0.5	55.7 ± 2.54	97.6 ± 1.26
EMOSAM	0.3 ± 0.18	0.31 ± 0.19	0.33 ± 0.27	0.56 ± 0.52	**0 ± 0**	84.5 ± 0.95	49.9 ± 0.3	98.9 ± 0.92
FABBOO SP	1.37 ± 0.42	1.61 ± 0.5	1.23 ± 0.73	9.76 ± 2.43	3.13 ± 0.64	82.8 ± 0.69	66.9 ± 0.37	89.5 ± 0.85
FABBOO EQOP	1.93 ± 0.83	2.28 ± 0.98	1.4 ± 0.42	9.07 ± 3.13	2.96 ± 0.85	82.8 ± 0.66	**67.1 ± 0.34**	89.3 ± 0.83

Table 16. Fairness and performance comparison across metrics on the *internet era* stream (one drift) with drift-aware scoring.

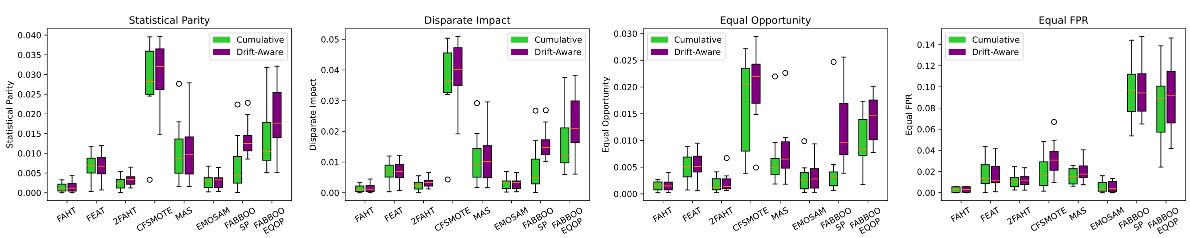

Fig. 17. Paired boxplots showing the differences between cumulatively and drift-awarely aggregated scores on the *internet era* stream with one drift.